

drained soil. Propagate by semiripe cuttings in late summer or fall.

M. aquisargentii. Evergreen, upright, densely leaved shrub. H and S 6ft (2m). Leaves are bright green, often with 3 leaflets, some oblong and finely toothed, others holly-shaped. Terminal clusters of berberislike, yellow flowers are sparsely produced in early summer. Z5–9

MAHONIA

BERBERIDACEAE

Genus of evergreen shrubs, grown for their foliage, their often fragrant, rounded, bell-shaped, yellow flowers, and, with the tall species and cultivars, for their deeply fissured bark. Large mahonias are good specimen plants; low-growing ones are excellent for groundcover. Fully hardy to frost tender. Prefers shade or partial shade and fertile, well-drained, but not too dry, soil. Propagate species by leaf-bud or semiripe cuttings in late summer to fall, or by seed in fall, selected forms by leaf-bud or semiripe cuttings only.

M. aquifolium (Oregon grape) illus. p.148. ‘**Apollo**’ is a wide-spreading, low, stem-rooting shrub. H 2ft (60cm), S 3ft (1m). Has dark-green, divided leaves, with up to 9 spiny-toothed leaflets, that turn bronze in winter. Deep golden yellow flowers are borne in large, pyramidal clusters in spring. Blue-black berries follow. ‘**Fascicularis**’ see *M. x wagneri* ‘Pinnacle’. Z5-9

M. bealei, syn. *M. japonica* Bealei Group. Erect shrub with stout, upright branches. H 6ft (2m), S 10ft (3m). Blue-green leaves are divided into broad leaflets. From late fall to early spring, bears fragrant, pale yellow flowers in short, upright racemes, followed by blue-purple berries. Z5–9

M. japonica illus. p.144. **Bealei Group** see *M. bealei*.

M. lomariifolia. See *M. oiwakensis* subsp. *lomariifolia*.

M. x media ‘Buckland’ illus. p.118. ‘**Charity**’ illus. p.118. ‘**Winter Sun**’ is an erect shrub. H to 15ft (5m), S to 12ft (4m). Has long leaves with 17–21 ovate to lance-shaped, spiny, dark green leaflets. Bears bright yellow flowers in densely clustered, arching racemes, from late fall to late winter. Z5–9

M. nepalensis ‘Cabaret’. See *M. nitens* ‘Cabaret’.

M. nitens ‘Cabaret’, syn. *M. nepaulensis* ‘Cabaret’. Small, compact, freely branching shrub. H 5ft (1.5m), S 4ft (1.2m). Glossy, ovate, dark green leaves have pairs of sparsely spined leaflets, each with a triangular red stain at the base, which is conspicuous in winter. Bears many small, deep yellow flowers with red outer petals from midsummer to early fall. Z6–9

M. oiwakensis subsp. lomariifolia, syn. *M. lomariifolia*. Evergreen, very upright shrub or small tree. H 10ft (3m), S 6ft (2m). Large, dark green leaves each have up to 41 narrow, hollylike, spiny leaflets. Fragrant, bright yellow flowers are produced in dense, upright spikes from early fall to winter. Z6–9

M. pinnata of gardens. See *M. x wagneri* ‘Pinnacle’.

M. repens. Evergreen, upright shrub, spreading by underground stems. H 1ft (30cm), S 6ft (2m). Blue-green leaves each

consist of wavy-edged, ovate leaflets, with bristlelike teeth. Dense clusters of deep yellow flowers are borne from mid- to late spring, followed by blue-black berries. Z4–8

M. ‘Undulata’. See *M. x wagneri* ‘Undulata’.

M. x wagneri ‘Pinnacle’, syn. *M. aquifolium* ‘Fascicularis’, *M. pinnata* of gardens. Upright, evergreen shrub. H and S 5ft (1.5m). Has bronze young leaves, maturing to bright green, with sharply toothed, dull to dark green leaflets. Dense clusters of yellow flowers open in spring, followed by white-frosted, blue-black berries. Z5–8. ‘**Undulata**’ (syn. *M. ‘Undulata’*) H and S 6ft (2m) has glossy, dark green leaves, each with hollylike, wavy-edged leaflets that turn red-purple in winter. Bears deep yellow flowers.

MAIANTHEMUM

SYN. SMILACINA

False Solomon’s seal, May lily

LILIACEAE/CONVALLARIACEAE

Genus of perennials with extensive, clump-forming or spreading rhizomes, grown for their sprays of often scented flowers. Useful as groundcover in woodlands and wild areas. Fully hardy to frost tender. Prefers a position in shade and humus-rich, moist, sandy, neutral to acidic soil. Propagate by seed in fall or division in early spring.

M. bifolium illus. p.348.

M. canadense. Vigorous, ground-cover, spreading perennial. H 10cm (4in), S indefinite. Has pairs of large, upright, ovate, glossy leaves. Slender stems bear sprays of small, white flowers in early summer, followed by green mottled berries that ripen to red. Z3–8

M. racemosum, syn. *Smilacina racemosa* (False spikenard), illus. p.223.

MAIHUENIA

CACTACEAE

Genus of slow-growing, summer-flowering, alpine cacti, clump-forming with age, with cylindrical or round stems. Half hardy. Requires dappled shade or full sun and well-drained soil. Protect from winter rain. Propagate by seed or stem cuttings in spring or summer.

M. poeppigii illus. p.490.

MALCOLMIA

BRASSICACEAE/CRUCIFERAE

Genus of spring- to fall-flowering annuals and perennials. Frost hardy. Grow in sun and in fertile, well-drained soil. Propagate by seed sown outdoors in spring, summer or early fall. Self-seeds freely.

M. maritima (Virginian stock) illus. p.304.

MALEPHORA

AIZOACEAE

Genus of prostrate to erect, woody-based, perennial succulents with semicylindrical or bluntly 3-angled leaves. Frost tender. Star-shaped flowers open from late summer to fall, in daytime. Needs sun and very well-drained soil. Propagate by seed or stem cuttings in spring or summer.

M. crocea illus. p.496.

MALOPE

MALVACEAE

Genus of annuals and perennials, grown for their showy flowers that are ideal for cutting. Borderline hardy to fully hardy. Grow in sun and in fertile, well-drained soil. Propagate by seed sown outdoors in spring. Self-seeds freely.

M. trifida (Annual mallow) illus. p.305.

MALUS

Apple, Crab apple

ROSACEAE

Genus of deciduous, mainly spring-flowering trees and shrubs, grown for their shallowly cup-shaped flowers, fruits, foliage or fall color. Crab apples may be used to make preserves. Fully hardy. Prefers full sun, but tolerates partial shade; grows in any but waterlogged soil. In winter, cut out dead or diseased wood and prune to maintain a balanced branch system. Propagate by budding in late summer or by grafting in midwinter. Trees are sometimes attacked by aphids, caterpillars, and red spider mite, and are susceptible to fireblight and apple scab.

M. ‘Adirondack’, syn. *M. ADMIRATION*. Slow-growing, deciduous, narrowly upright tree. H 12ft (4m), S 6ft (2m). Has ovate, leathery, dark green leaves. Carmine-red buds open to red-tinged, white flowers in mid- to late spring, followed by small, long-lasting, roundish, red fruits on red stalks. Z4–8

M. ADMIRATION. See *M. ‘Adirondack’*.

M. baccata (Siberian crab apple). Deciduous, rounded tree. H and S 50ft (15m). Has ovate, dark green leaves and produces a profusion of single, white flowers from mid- to late spring, followed by tiny, rounded, red or yellow crab apples in fall. H6. **var. mandschurica** (Manchurian crab apple) illus. p.69.

M. brevipes ‘Wedding Bouquet’. Deciduous, compact, wide-spreading tree or shrub. H 10–12ft (3–4m), S 12–15ft (4–5m). Has ovate to oblong, finely toothed, dark green leaves. Abundant clusters of saucer-shaped, pure white flowers are borne in late spring, followed by roundish, bright red fruits, which last into winter. Z4–8

M. ‘Butterball’ illus. p.92.

M. ‘Comtesse de Paris’. Small, broadly conical tree. H 15ft (5m), S 12ft (4m). Has variable, wavy-margined, occasionally lobed, mid-green leaves. Pure white flowers open from pink buds in mid- to late spring, followed by long-stalked, small, lemon-shaped, clear yellow fruits that remain on branches to mid-winter. Z4–8

M. bhutanica, syn. *M. toringoides*. Deciduous, spreading tree. H 25ft (8m), S 30ft (10m). Ovate to lance-shaped, usually deeply lobed, glossy, mid-green leaves turn yellow in fall. Bears single, white flowers in late spring, and rounded or egg-shaped, red-flushed, yellow crab apples in fall. Z4–8

M. coronaria ‘Charlottae’. Spreading tree. H and S 28ft (9m). Broadly ovate, lobed or deeply toothed leaves are tinged red when young, then dark green, and turning red and orange in fall. Semidouble, pale pink flowers open in late spring. Z4–8

M. ‘Cowichan’ illus. p.90.

MALUS

M. ‘Evereste’. Conical tree. H 22ft (7m), S 20ft (6m). Has ovate, sometimes lobed, dark green leaves. White flowers are freely borne from red buds in late spring. Red-flushed, orange-yellow fruits follow. Z4–8

M. x floribunda (Japanese crab apple) illus. p.84.

M. hupehensis (Hupeh crab) illus. p.69.

M. JELLY ING (‘Mattfru’). Compact, round-headed, small tree. H and S 12ft (4m). Has broadly ovate, mid-green leaves. Produces abundant, fragrant, pure white flowers in mid- to late spring. The attractive, orange-pink, roundish fruits persist until early winter. Self-fertile; used as a universal pollinator for orchard and other crab apples. Disease resistant and fruits are good for making jelly. Z4–8

M. ‘John Downie’ illus. p.91.

M. ‘Laura’. Slow-growing, columnar tree with erect branches. H 6ft (2m), S 2ft (0.6m). Has ovate, occasionally lobed leaves, dark purple when young, turning dark reddish-green with red veins by midsummer. Bears pink and white flowers in mid-spring, followed by long-lasting, large, roundish, maroon-red fruits. Self-fertile. Ideal for containers; fruits make good jelly. Z4–8

M. x moerlandsii ‘Profusion’. See *M. ‘Profusion’*.

M. ‘Neville Copeman’, syn. *M. x purpurea* ‘Neville Copeman’. Spreading tree. H and S 28ft (9m). Ovate, dark green leaves are purplish-red when young. Single, dark purplish-pink flowers are borne from mid- to late spring; these are followed by rounded, orange-red to carmine crab apples. Z4–8

M. niedzwetskyana, syn. *M. pumila* var. *niedzwetskyana*. Spreading tree. H 20ft (6m), S 25ft (8m). Ovate leaves are red when young, later purple. Produces clusters of single, deep reddish-purple flowers in late spring, then very large, conical, reddish-purple crab apples. Z4–8

M. prattii. Upright when young, later spreading tree. H and S 30ft (10m). Ovate, red-stalked, glossy, mid-green leaves become orange and red in fall. Single, white flowers in late spring are followed by small, rounded or egg-shaped, white-flecked, red crab apples. Z4–8

M. ‘Princeton Cardinal’ Vigorous, spreading, small, tree. H and S 15–20ft (5–6m). Has broadly ovate, glossy leaves, purplish-red when young, maturing to dark green, reddish beneath. Bears vivid, pinkish-red flowers in early spring, followed by small, round, purple fruits. Disease resistant. Z4–8

M. ‘Professor Sprenger’ illus. p.91.

M. ‘Profusion’, syn. *M. x moerlandsii* ‘Profusion’, illus. p.71.

M. pumila var. niedzwetskyana. See *M. niedzwetskyana*.

M. x purpurea ‘Neville Copeman.’ See *M. ‘Neville Copeman’*.

M. ‘Red Jade’. Weeping tree. H 12ft (4m), S 20ft (6m). Ovate leaves are dark green. In late spring, bears single, white flowers, sometimes pale pink-flushed, then long-lasting, rounded to egg-shaped, red crab apples. Z4–8

M. x robusta. Vigorous, deciduous, spreading tree. H 40ft (12m), S 30ft (10m). Has ovate, dark green leaves. Bears profuse, single, white or pink flowers